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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 312 — Past Questions & Materials
by Renance

Course: MEE 312

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 300

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Paper 1 of 3 — MEE 312 Mechanics of Machines

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF TECHNOLOGY @ ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2024/2025 Session

urse & Title: MEE 312 Mw

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Time Allowed: 2 Hours

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What distinguishes a

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Determine the degre

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Paper 2 of 3 — MEE 312 mechanics of machines 2025

Level 300 • First Semester • PastQuestion

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truction; Answer four (4) questions.

ESTION ONE

What distinguishes a machine from a structure?

Determine the degrees of freedom for the specified mechanism in Figure 1.

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ESTION TWO

Illustrate four-bar chain inversions with sketches.

State Grashof's law and its limitations.

ESTION THREE

Define: (i) Degree of freedom, (ii) Kinematic chain, (iii) Higher pair, {iv)Quaternary joint.

List and explain ten (10) gear tern'inologies or nomenclatures with a good sketch.

ESTION FOUR

What is a Cam? Identify five applications of Cams.

For the mechanism shown in the "igure, calculate its degrees of freedom (F).

ESTION FIVE

+ gearing of a machine tool is shown in Fig. 1. The motor shaft is connected to gear A and rotates 175 r.p.m. The gear wheels B, C, D and E are fixed to parallel shafts rotating together. The final gear F is

don the output shaft. What is the speed of gear F? The number of teeth on each gear is given below.

Gear __ A B Cc D E F

Number or oe

Teeth

76 26 65

Paper 3 of 3 — MEE 312: THEORY OF MACHINES I GEARS & GEARED SYSTEMS

Level 300 • First Semester • Handout

GEARS & GEARED SYSTEMS

MEE 312: THEORY OF MACHINES I

LECTURER: DR C.O. MGBEMENA

Introduction to Gear

- Gears are machine elements that transmit motion by means of successively engaging teeth.
- The gear teeth act like small levers.
- Gears are highly efficient (nearly 95%) due to primarily rolling contact between the teeth, thus the motion transmitted is considered as positive.
- Gears essentially allow positive engagement between teeth so high forces can be transmitted while still undergoing essentially rolling contact.
- Gears do not depend on friction and do best when friction is minimized.

Advantages and Disadvantages of Gear Drive

The following are the advantages and disadvantages of the gear drive as compared to belt, rope and chain drives :

Advantages:

1. It transmits exact velocity ratio.
2. It may be used to transmit large power.
3. It has high efficiency.
4. It has reliable service.
5. It has compact layout.

Disadvantages:

1. The manufacture of gears requires special tools and equipment.
2. The error in cutting teeth may cause vibrations and noise during operation.

Classification of Toothed Wheels

Gears may be classified according to the relative position of the axes of revolution.

The axes may be

1. Gears for connecting parallel shafts,

2. Gears for connecting intersecting shafts,
3. Gears for neither parallel nor intersecting shafts.

Gears for connecting parallel shafts

- Spur gears: Spur gears are the most common type of gears. They have straight teeth and are mounted on parallel shafts. Sometimes, many spur gears are used at once to create very large gear reductions.

□ Spur gears are the most commonly used gear type. They are characterised by teeth, which are perpendicular to the face of the gear.

□ Spur gears are most commonly available and are generally the least expensive.

- Limitations: Spur gears generally cannot be used when a direction change between the two shafts is required.

- Advantages: Spur gears are easy to find, inexpensive, and efficient

- Parallel helical gears: The teeth on helical gears are cut at an angle to the face of the gear. When two teeth on a helical gear system engage, the contact starts at one end of the tooth and gradually spreads as the gears rotate, until the two teeth fully engage.

Helical gears have the following differences from spur gears of the same size:

□ Tooth strength is greater because the teeth are longer,

□ Greater surface contact on the teeth allows a helical gear to carry more load than a spur gear

□ The longer surface of contact reduces the efficiency of a helical gear relative to a spur gear

Rack and pinion

- The rack is like a gear whose axis is at infinity mathematically but practically a gear of larger length.

□ Racks are straight gears that are used to convert rotational motion to translational motion using a gear mesh. (They are in theory a gear with an infinite pitch diameter).

□ In theory, the torque and angular velocity of the pinion gear are related to the Force and the velocity of the

rack by the radius of the pinion gear, as is shown.

□ Perhaps the most well-known application of a rack is the rack and pinion steering system used on many cars in the past.

Gears for connecting intersecting shafts

- Bevel gears are useful when the direction of a shaft's rotation needs to be changed.
- They are usually mounted on shafts that are 90 degrees apart but can be designed to work at other angles as well.
- The teeth on bevel gears can be straight, spiral or hypoid.

□ Straight bevel gear teeth have the same problem as straight spur gear teeth, as each tooth engages; it impacts the corresponding tooth all at once.

□ Just like with spur gears, the solution to this problem is to curve the gear teeth.

□ These spiral teeth engage just like helical teeth: the contact starts at one end of the gear and progressively spreads across the whole tooth.

Neither parallel nor intersecting shafts

Helical gears may be used to mesh two shafts that are not parallel, although they are still primarily used in parallel shaft applications.

A special application in which helical gears are used is a crossed gear mesh, in which the two shafts are perpendicular to each other.

Worm and worm gear: Worm gears are used when large gear reductions are needed. It is common for worm gears to have reductions of 20:1, and even up to 300:1 or greater.

Many worm gears have an interesting property that no other gear set has: the worm can easily turn the gear, but the gear cannot turn the worm.

This is because the angle on the worm is so shallow that when the gear tries to spin it, the friction between the gear and the worm holds the worm in place.

This feature is useful for machines such as conveyor systems, in which the locking feature can act as a brake for the conveyor when the motor is not turning.

One other very interesting usage of worm gears is in the

Torsen differential, which is used on some high-performance cars and trucks.

Gear Nomenclature/Terminology

Gear

Terminologies/Nomenclature...

cont'd

- Addendum: The radial distance between the Pitch Circle and the top of the teeth.
- Dedendum: The radial distance between the bottom of the tooth to the pitch circle.
- Base Circle: The circle from which is generated the involute curve upon which the tooth profile is based.
- Center Distance: The distance between the centers of two gears.
- Circular Pitch: Millimeter of Pitch Circle circumference per tooth.
- Circular Thickness: The thickness of the tooth measured along an arc following the Pitch Circle
- Clearance: The distance between the top of a tooth and the bottom of the space into which it fits on the meshing gear.
- Contact Ratio: The ratio of the length of the Arc of Action to the Circular Pitch.
- Diametral Pitch: Teeth per mm of diameter.
- Face: The working surface of a gear tooth, located between the pitch diameter and the top of the tooth.
- Face Width: The width of the tooth measured parallel to the gear axis.
- Flank: The working surface of a gear tooth, located between the pitch diameter and the bottom of the teeth
- Gear: The larger of two meshed gears. If both gears are the same size, they are both called "gears".
- Land: The top surface of the tooth.
- Line of Action: That line along which the point of contact between gear teeth travels, between the first point of contact and the last.
- Module: Millimeter of Pitch Diameter to Teeth.
- Pinion: The smaller of two meshed gears.
- Pitch Circle: The circle, the radius of which is equal to the distance from the center of the gear to the pitch point.
- Diametral pitch: Teeth per millimeter of pitch diameter.
- Pitch Point: The point of tangency of the pitch circles of two meshing gears, where the Line of Centers crosses the pitch circles.

- Pressure Angle: Angle between the Line of Action and a line perpendicular to the Line of Centers.
- Root Circle: The circle that passes through the bottom of the tooth spaces.
- Working Depth: The depth to which a tooth extends into the space between teeth on the mating gear.

Practice Problem/Examples